



Online Examination System — Complete Project Flow

React + Vite

React Router v6

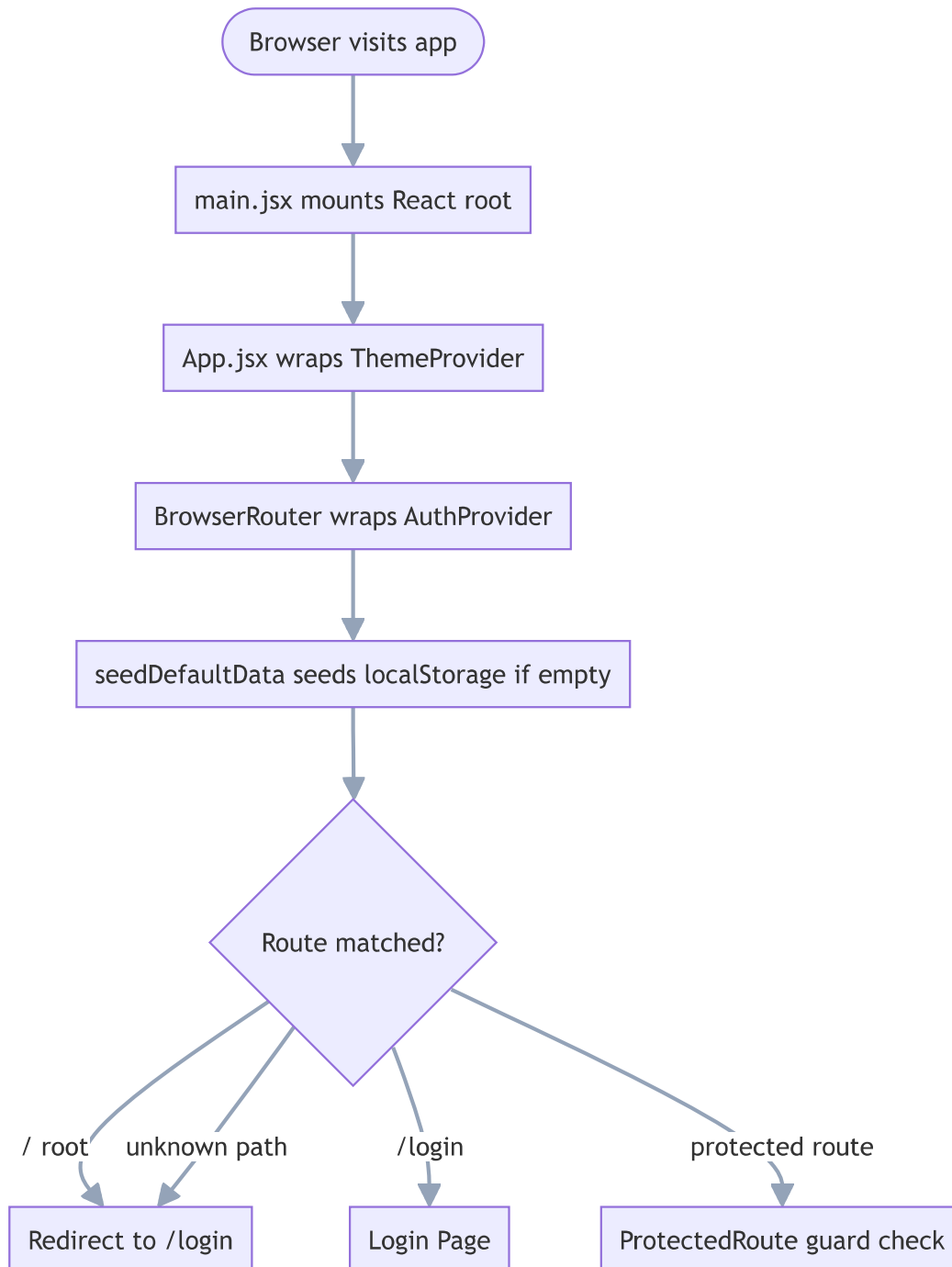
Context API

localStorage

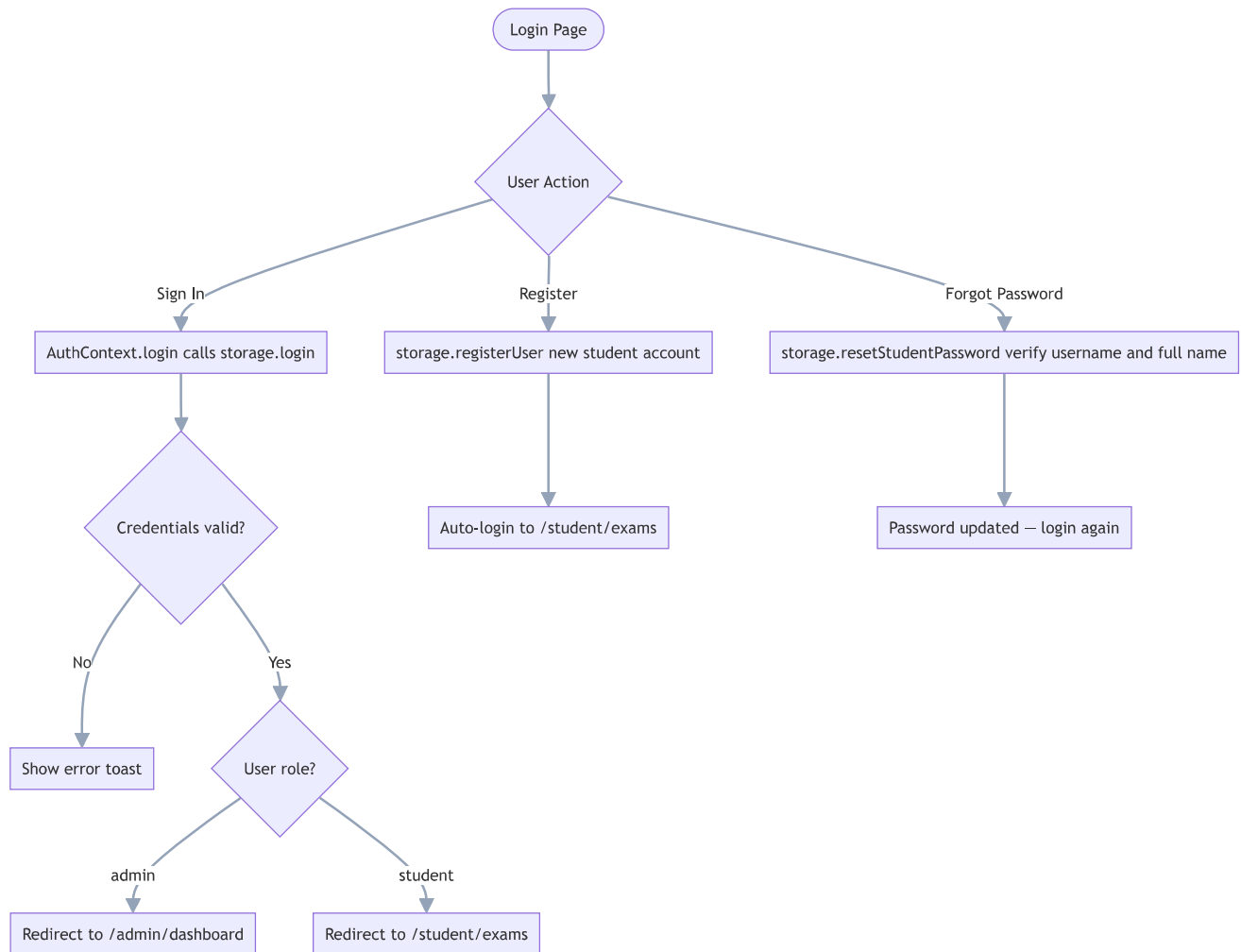
Tailwind CSS



1 Application Bootstrap



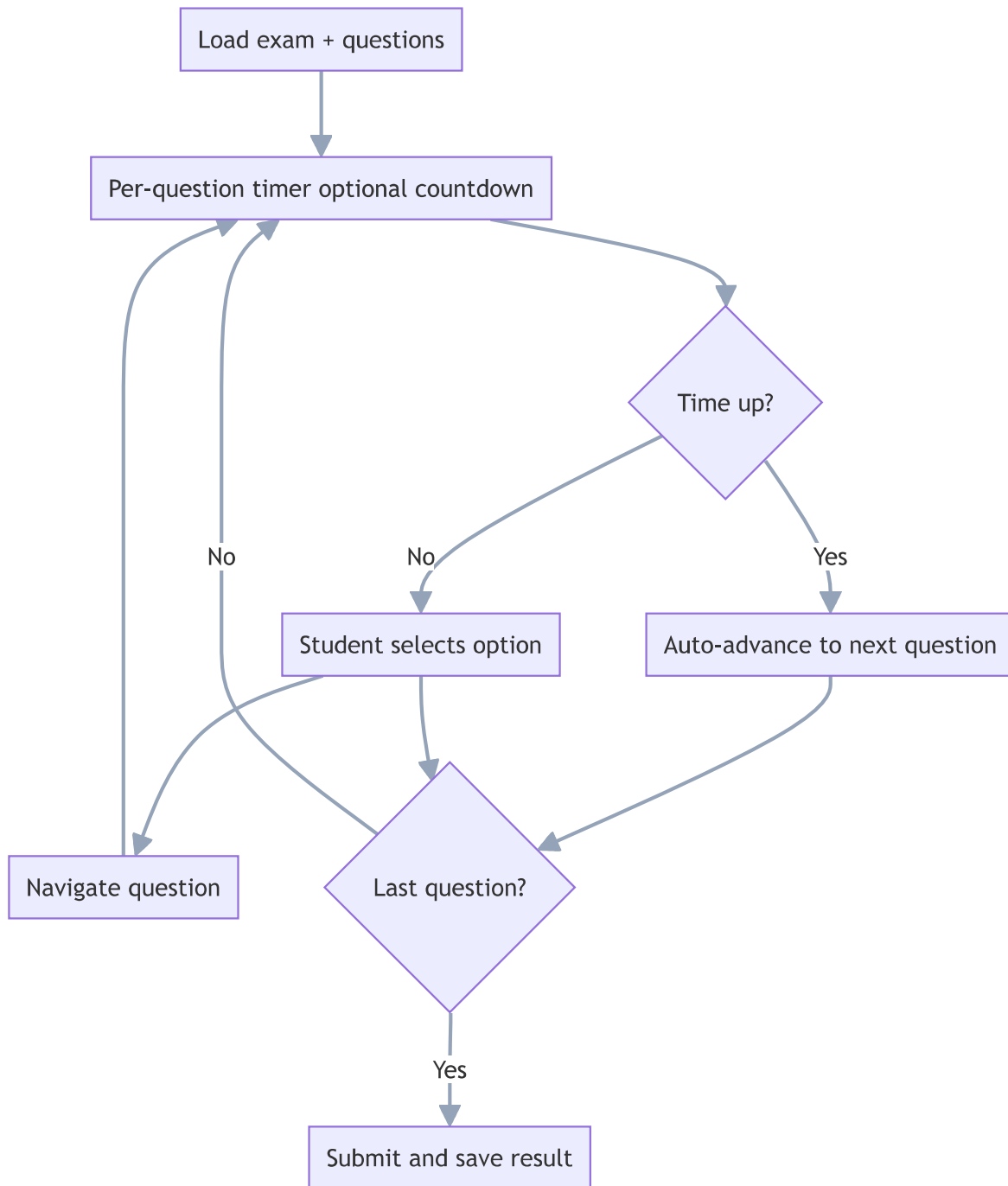
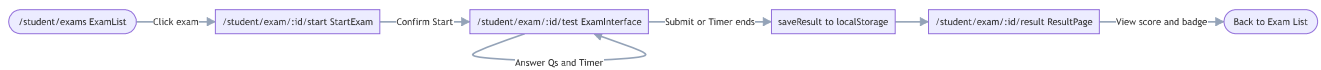
2 Authentication Flow



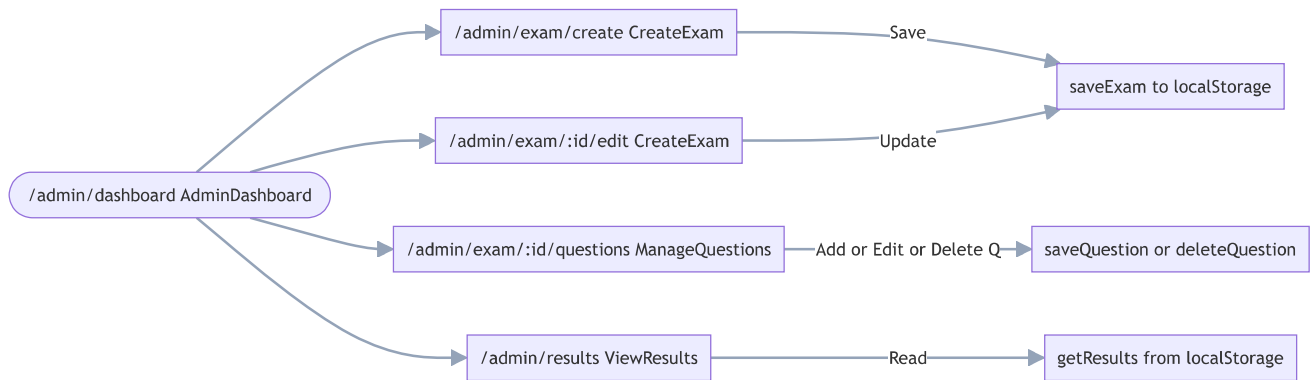
3 ProtectedRoute Guard



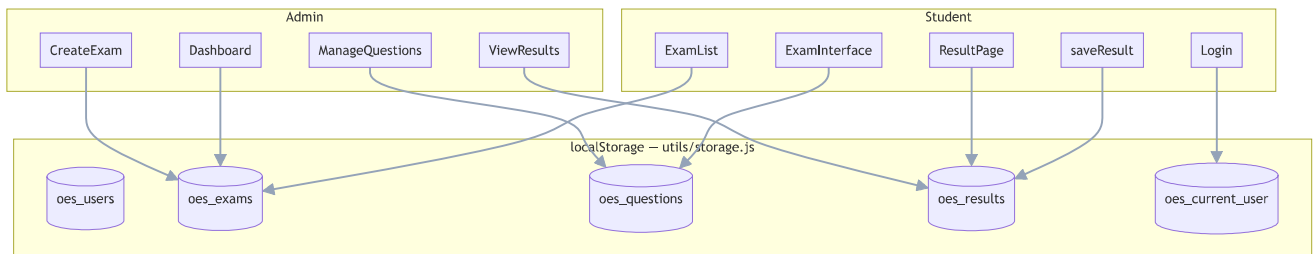
4 Student Exam Flow



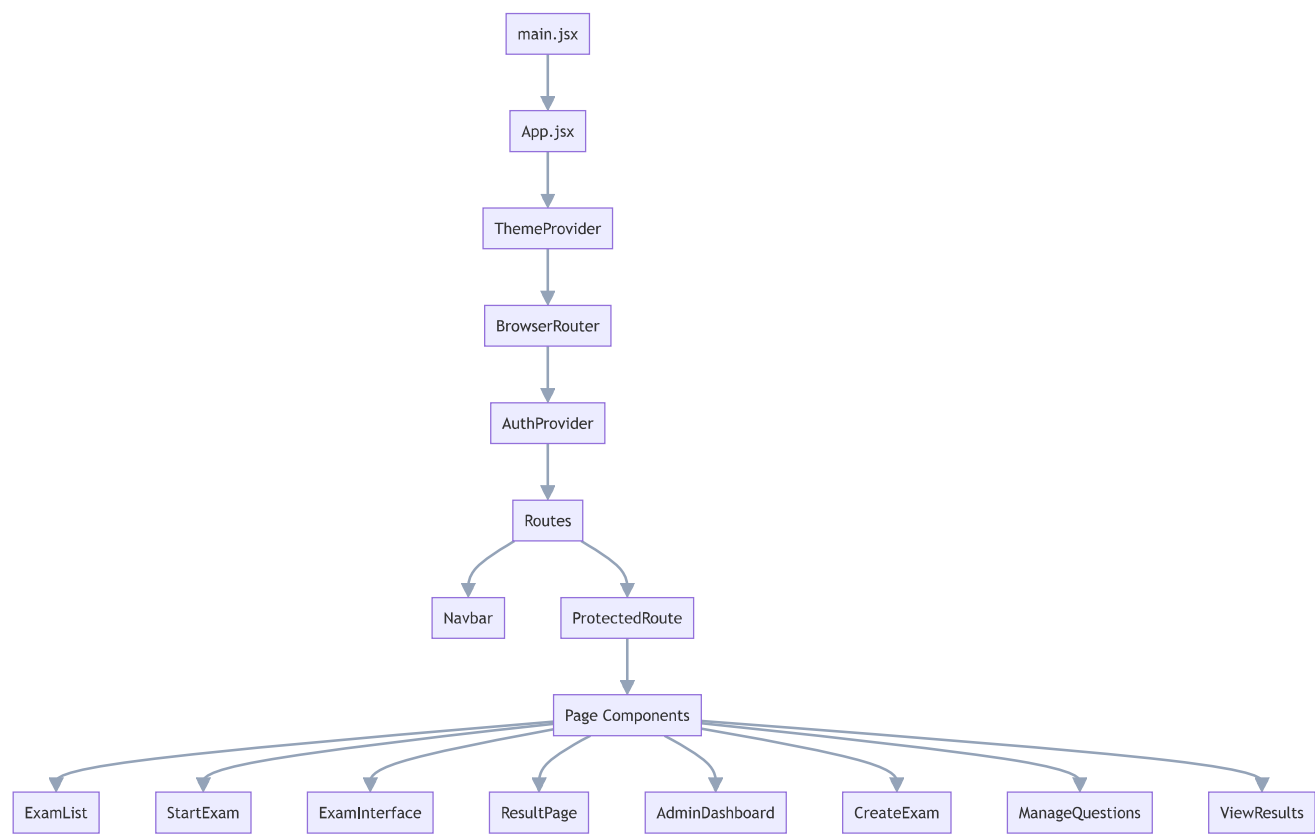
5 Admin Flow



6 Data Layer — localStorage



7 Component Hierarchy



8 Context / Global State

Context	Provides	Used By
AuthContext	user , login() , logout() , loading	ProtectedRoute, Navbar, all pages
ThemeContext	theme , toggleTheme()	Navbar, global CSS

9 Full Route Map

Path	Component	Role Required
/	→ redirect /login	Public
/login	Login	Public
/student/exams	ExamList	student
/student/exam/:id/start	StartExam	student
/student/exam/:id/test	ExamInterface	student
/student/exam/:id/result	ResultPage	student
/admin/dashboard	AdminDashboard	admin
/admin/exam/create	CreateExam	admin
/admin/exam/:id/edit	CreateExam	admin
/admin/exam/:id/questions	ManageQuestions	admin
/admin/results	ViewResults	admin
*	→ redirect /login	Public